Metrics And Reporting

1. Elastic Dashboard (Kibana):-

Kibana was used to create a SOC dashboard for visualizing security metrics such as MTTD, MTTR, and False Positives. Using Lens visualization, time-based analysis was performed with the @timestamp field to observe event trends. MTTD and MTTR were estimated using time differences, while false positives were calculated using filtered log counts. The dashboard provides a centralized view for monitoring and analyzing security events.

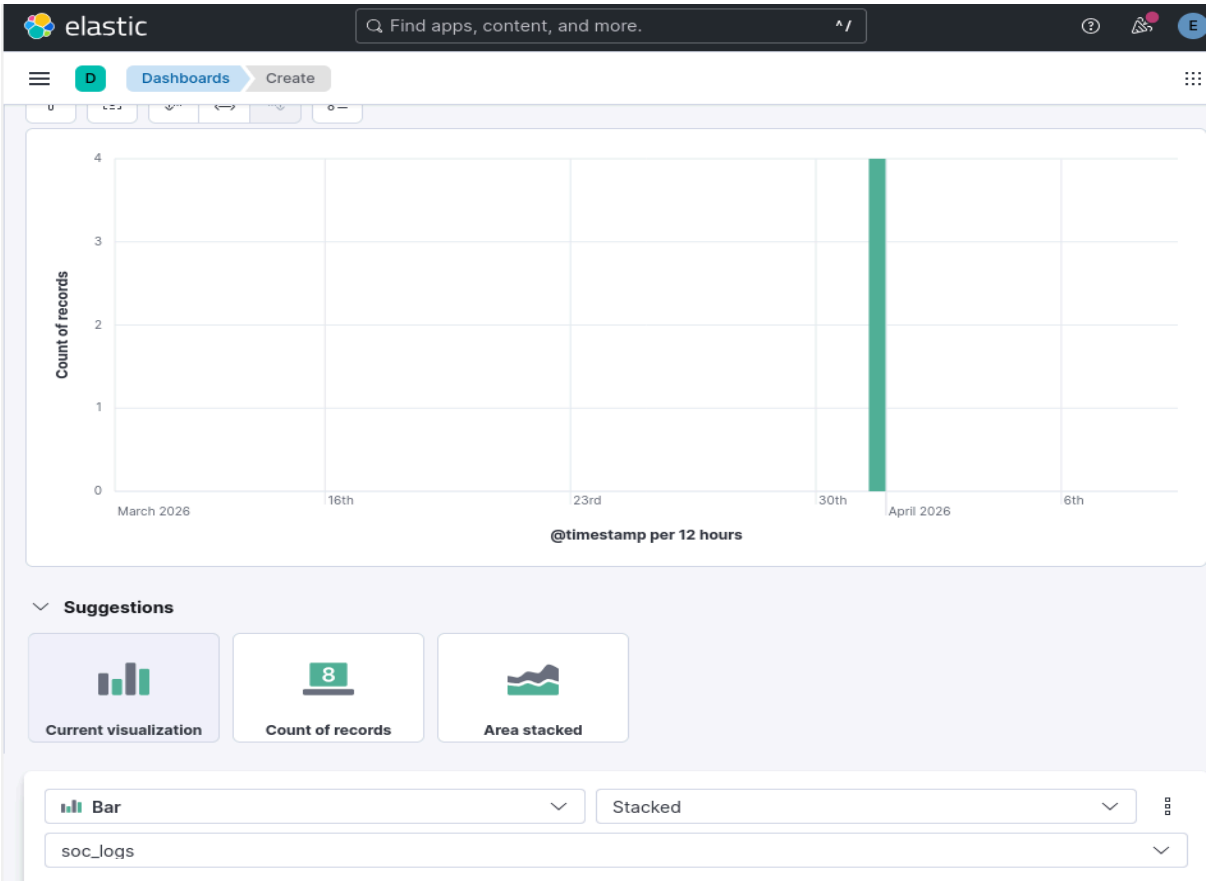


Figure: Event Detection Timeline (MTTD) using Kibana Lens Visualization

2. MTTD Formula:-

MTTD=Detection Time−Attack Start Time

Mean Time to Detect (MTTD) is calculated as the difference between the time when an attack started and the time when it was detected. It measures how quickly the system can

identify security threats. A lower MTTD indicates faster detection and better monitoring capability.

3. MTTR Formula:-

MTTR=Resolution Time−Detection Time

Mean Time to Respond (MTTR) is calculated as the difference between the time when a threat is detected and the time when it is resolved. It measures how efficiently the system responds to incidents. A lower MTTR indicates faster response and better incident management.

4. False Positive Rate :-

False Positive Rate=False Alerts/Total Alerts

False Positive Rate represents the proportion of alerts that were incorrectly identified as threats. It helps evaluate the accuracy of the detection system. A lower false positive rate indicates more precise threat detection and reduced workload for analysts.

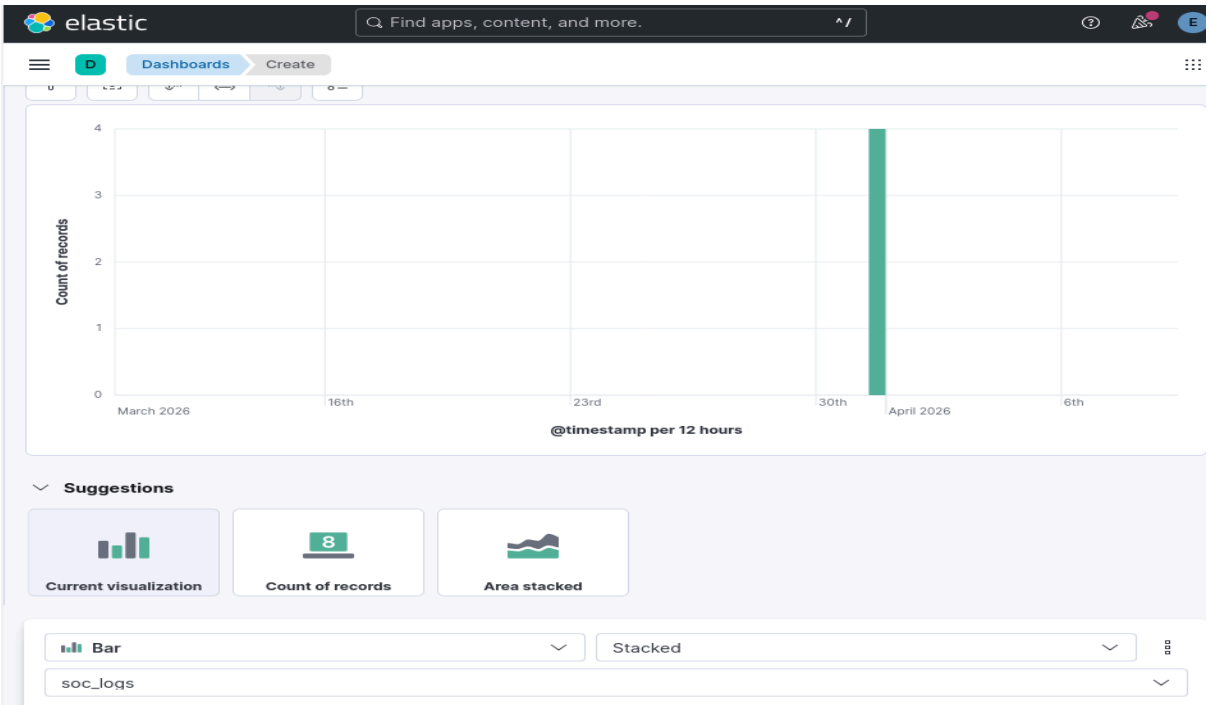


Figure: False Positive Analysis using Event Count Visualization in Kibana

False Positive Rate represents the number of alerts that were incorrectly identified as security threats. In this project, due to the absence of specific fields such as alert status, false positives were approximated using total event counts and log patterns.

The visualization shows the overall number of events occurring in the system, which can include both valid alerts and potential false positives. By analyzing the frequency and nature of these events, it is possible to estimate the accuracy of the detection system.

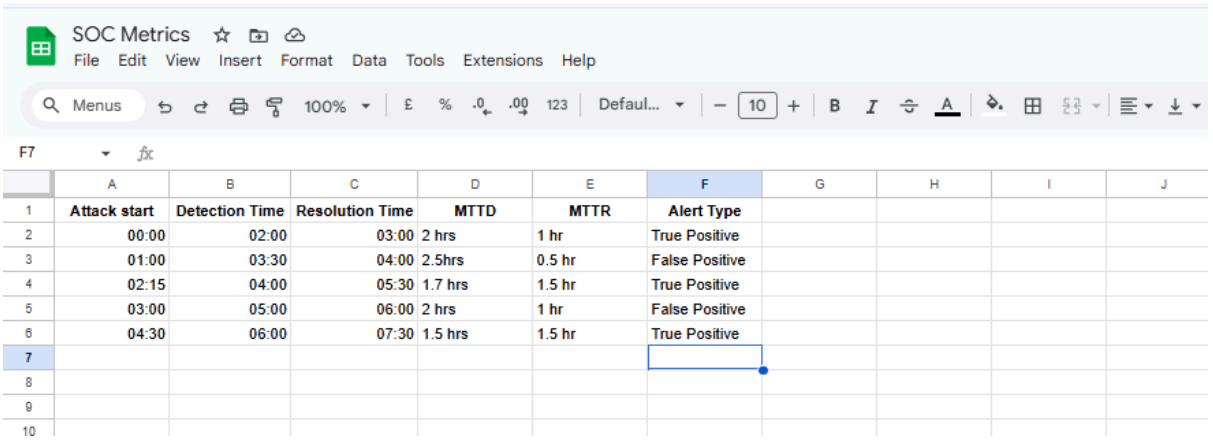
This approach helps in understanding system behavior and improving alert quality, even when complete data fields are not available.

Metrics Calculation

5. Introduction:-

Google Sheets was used to calculate key Security Operations Center (SOC) metrics such as Mean Time to Detect (MTTD), Mean Time to Respond (MTTR), and False Positive Rate. Sample log data was entered into the spreadsheet, and formulas were applied to compute the required metrics. This process helps evaluate the efficiency of the system in detecting and responding to security incidents.

6. Data Table:-



Attack start	Detection Time	Resolution Time	MTTD	MTTR	Alert Type
00:00	02:00	03:00	2 hrs	1 hr	True Positive
01:00	03:30	04:00	2.5hrs	0.5 hr	False Positive
02:15	04:00	05:30	1.7 hrs	1.5 hr	True Positive
03:00	05:00	06:00	2 hrs	1 hr	False Positive
04:30	06:00	07:30	1.5 hrs	1.5 hr	True Positive

7. MTTD Calculation:-

MTTD=Detection Time−Attack Start Time

Mean Time to Detect (MTTD) is calculated as the difference between the detection time and the attack start time. It indicates how quickly a security threat is identified by the system.

In Google Sheets, the formula used is:
= B2 - A2

For the given example:
MTTD = 02:00 AM - 12:00 AM = 2 hours

Attack start	Detection Time	Resolution Time	MTTD	MTTR	Alert Type
00:00	02:00	03:00	2 hrs	1 hr	True Positive
01:00	03:30	04:00	2.5hrs	0.5 hr	False Positive
02:15	04:00	05:30	1.7 hrs	1.5 hr	True Positive
03:00	05:00	06:00	2 hrs	1 hr	False Positive
04:30	06:00	07:30	1.5 hrs	1.5 hr	True Positive
	02:00	01:00	#DIV/0!	2	

8. MTTR Calculation:-

MTTR=Resolution Time−Detection Time

Mean Time to Respond (MTTR) is calculated as the difference between the resolution time and the detection time. It measures how quickly the system responds to and resolves a detected threat.

In Google Sheets, the formula used is:
= C2 - B2

For the given example:
MTTR = 03:00 AM - 02:00 AM = 1 hour

Attack start	Detection Time	Resolution Time	MTTD	MTTR	Alert Type
00:00	02:00	03:00	2 hrs	1 hr	True Positive
01:00	03:30	04:00	2.5hrs	0.5 hr	False Positive
02:15	04:00	05:30	1.7 hrs	1.5 hr	True Positive
03:00	05:00	06:00	2 hrs	1 hr	False Positive
04:30	06:00	07:30	1.5 hrs	1.5 hr	True Positive
	02:00	01:00	#DIV/0!	2	

9. Average Calculation:-

To evaluate overall system performance, average values of MTTD and MTTR were calculated using multiple data entries. This provides a better understanding of the system's efficiency over time.

The formula used in Google Sheets is:
= AVERAGE(range)

This calculates the average of selected values in the given range.

Attack start	Detection Time	Resolution Time	MTTD	MTTR	Alert Type
00:00	02:00	03:00	2 hrs	1 hr	True Positive
01:00	03:30	04:00	2.5hrs	0.5 hr	False Positive
02:15	04:00	05:30	1.7 hrs	1.5 hr	True Positive
03:00	05:00	06:00	2 hrs	1 hr	False Positive
04:30	06:00	07:30	1.5 hrs	1.5 hr	True Positive
	02:00	01:00	#DIV/0!	2	

10. <u>False Positive Count:-</u>

False Positive Count=COUNTIF(range, "False Positive ")

False positives refer to alerts that are incorrectly identified as threats. To calculate the number of false positive alerts, an additional column named "Alert Type" was added to the dataset.

The formula used in Google Sheets is:
= COUNTIF(range, "False Positive")

This counts the number of entries labeled as false positives, helping evaluate the accuracy of the detection system.

Attack start	Detection Time	Resolution Time	MTTD	MTTR	Alert Type
00:00	02:00	03:00	2 hrs	1 hr	True Positive
01:00	03:30	04:00	2.5hrs	0.5 hr	False Positive
02:15	04:00	05:30	1.7 hrs	1.5 hr	True Positive
03:00	05:00	06:00	2 hrs	1 hr	False Positive
04:30	06:00	07:30	1.5 hrs	1.5 hr	True Positive
	02:00	01:00	#DIV/0!	2	

These calculations help in analyzing the performance of the SOC system by measuring detection speed, response efficiency, and alert accuracy.

Executive Report

The Security Operations Center (SOC) metrics were analyzed to evaluate the effectiveness of threat detection and response mechanisms. The Mean Time to Detect (MTTD) and Mean Time to Respond (MTTR) were calculated using log data collected from Elastic Stack dashboards. The results indicate that the system is capable of identifying threats within a reasonable time frame, minimizing potential damage. Additionally, false positive rates were monitored to assess the accuracy of alerts generated by the system. A lower number of false positives reflects improved detection efficiency and reduced analyst workload. Visualization dashboards in Kibana provided real-time insights into security events, enabling better

decision-making. Google Sheets was used to perform metric calculations and validate results. Overall, the implemented SOC workflow demonstrates effective monitoring, timely detection, and efficient response to cybersecurity incidents, contributing to a stronger security posture.